

REUBEN SAMSON RAJ

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CAREER SUMMARY

Ph.D. candidate in Computer Science specializing in programmable network security for Smart-Grid/ICS/OT with 8+ publications in top-tier IEEE/ACM venues and Best Paper Award recognition. Combines research expertise in programmable networks, security frameworks with industry experience in cloud orchestration, 5G networks, and software validation.

EDUCATION

Ph.D. in Computer Science, University of Arkansas, Fayetteville, AR Aug 2021 – Dec 2026

M.S. in Computer Engineering, University of Arkansas, Fayetteville, AR Aug 2021 – Dec 2024

B.E. in Electronics and Communication Engineering, College of Engineering, Guindy, Anna University, Chennai Aug 2006 – June 2010

SKILLS

Languages	: Python, P4(Tofino, bmv2), C, Bash
Networking	: SDN, P4, 5G Core, EPC, TCP/IP, Mininet, Wireshark, iperf
Cloud & Orchestration	: Kubernetes, Docker, OpenStack
Security	: ICS/OT (Modbus, DNP3) Network Security, IDA Pro, PE Studio
Tools	: Linux, Git, FastAPI, OpenCV, NVIDIA GPU Stack

WORK EXPERIENCE

University of Arkansas, Fayetteville, AR

Graduate Research Assistant Aug 2021 – Present

- Independently research on secure Smart Grid/ICS networks using P4-programmable switches (Tofino, bmv2)
- Act as the go-to person for P4-related issues, including Tofino setup, coding and advising lab peers on technical solutions
- Mentor and provide research directions for undergrads on thesis projects
- Recognized by advisor for research independence, peer-level contributions, and leadership

Nokia Bell Labs, New Providence, NJ

Network Architecture Intern Jun 2023 – Aug 2023

- Deployed Kubernetes clusters on Nokia Airframe servers, simulating multi-cloud setups
- Optimized YoLov7-based object detection by transforming it into a FastAPI REST service, reducing RTT latency from 500 ms to 30 ms
- Integrated NVIDIA GPU plug-ins with Kubernetes to enhance GPU utilization for inference pods
- Scaled the video inference system, enabling real-time webcam feed processing with OpenCV

Cisco Systems, Bangalore

Software Engineer Jun 2020 – Jul 2021

- Conducted integration and end-to-end testing for Cisco's 5G Core Network Functions – SMF, UPF, SGW
- Deployed and managed Kubernetes-based software deployments in production environments
- Uncovered several critical bugs on RADIUS interface

Parallel Wireless, Bangalore

Member of Technical Staff Oct 2018 – May 2020

- Performed QA validation for Parallel Wireless' gateway products – TWAG (WiFi gateway) and SAEGW
- Validated different call models in TWAG – SIM based, portal authentication, end-to-end qualification
- Developed Python automation for SAEGW Gx-interface tests; reduced manual test execution time by ~60%

Nokia Networks, Bangalore
R & D Engineer

Apr 2015 – Oct 2018

- Software validation for Nokia's Home Subscriber Server (HSS) and 5G UDM Network Function
- Hands on experience with VNF bring up on OpenStack, Nokia CBAM (Cloud Band Application Manager)
- Led test strategy for HSS/5G UDM VNFs, improving defect detection before customer deployments and reducing field issues

Cisco Systems, Bangalore
Software Engineer

Apr 2012 – Apr 2015

- Performed software verification on Cisco StarOS ASR 5000, ASR 5500 SGSN gateway nodes
- Developed and maintained automated regression test suites for SGSN features using TCL scripting, enhancing Cisco's test automation framework for improved efficiency

SELECTED PUBLICATIONS

1. **Reuben Samson Raj** and Dong Jin. *A Framework to Evaluate PMU Networks for Resiliency Under Network Failure Conditions*, IEEE SmartGridComm 2022
2. **Reuben Samson Raj** and Dong Jin. *Dynamic Data Driven Security Framework for Industrial Control Networks using Programmable Switches*. DDDAS 2024
3. **Reuben Samson Raj** and Dong Jin. *Leveraging Data Plane Programmability Towards a Policy-driven In-Network Security Framework for Industrial Control Systems*. ACM e-Energy 2025
4. Sarvesh Bidkar, Rakesh Abbireddy, Maryam Amiri, **Reuben Samson Raj**, Anil Rana, Jeff McLaird, Paul Rea, Meryem Simsek and Jesse E. Simsarian. *Demonstration of Multi-Provider Network and Cloud Service Provisioning with Blockchain Smart Contracts*. ECOC 2025
5. **Reuben Samson Raj**, Dong Jin. *A Dynamic Data-Driven Framework for Enhancing Energy System Security and Resilience through Programmable Networks*. Book Chapter - DDDAS Handbook Vol. IV, Springer (Under Production)
6. Zhiyao He*, **Reuben Samson Raj***, Yanfeng Qu, Dong Jin. *Leveraging Programmable Switches towards Machine Learning-based Anomalous Event Classification of PMU data*. IEEE ICC 2026 (Accepted, * - Equal Contributions)

AWARDS AND HONORS

- **2nd place** - *Best Paper Award*, DDDAS 2024
- *Reginald R. Barney & Jameson A. Baxter Graduate Scholarship* (2023, 2022)
- *Department of Electrical Engineering and Computer Science Scholarship* (2023)
- *You Amaze and You Inspire* awards (2020), *CAP Award* (2013), Cisco
- *Best Feature Team Award* (2016), Nokia

ACADEMIC SERVICE & VOLUNTEERING

- Journal Reviewer, *ACM TOMACS*, 2025, 2026
- Conference Reviewer, *IEEE SmartGridComm*, 2026
- In charge of technical liaison with coding platform vendor, testing and onboarding group of volunteer judges